

REV. 00



PLAN REBUILD CENTER

ENGINE ASS'Y

RECEIVING INSPECTION CHECKSHEET

KOMATSU ENGINE SA12V140-1

MACHINE MODEL

WA 800-3

WO NO:

DISASSEMBLY SECTION

RECEIVING INSPECTION SHEET ENGINE

MACHINE MODEL :	WA 800-3	REPORT NO. :		DOCUMENT NO. :	RC/QR01/EG/01/05
ENGINE MODEL :	SA12V140-1	JOB ORDER NO. :		PUBL. DATE :	
ENGINE NO. :		DELIVERY DATE :		REVISION NO. :	
CUSTOMER NAME :		PAGE :			1 of 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

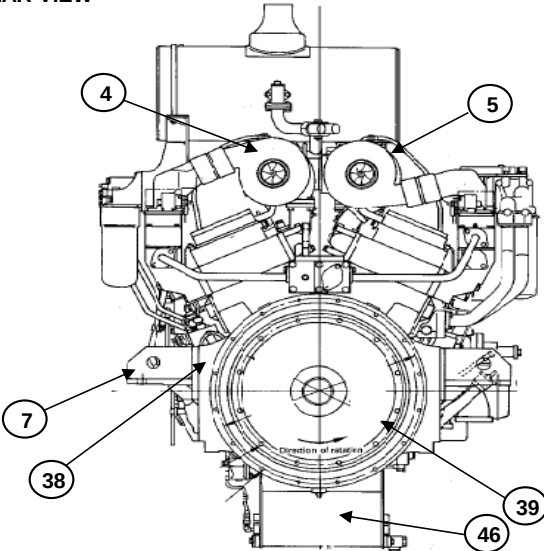
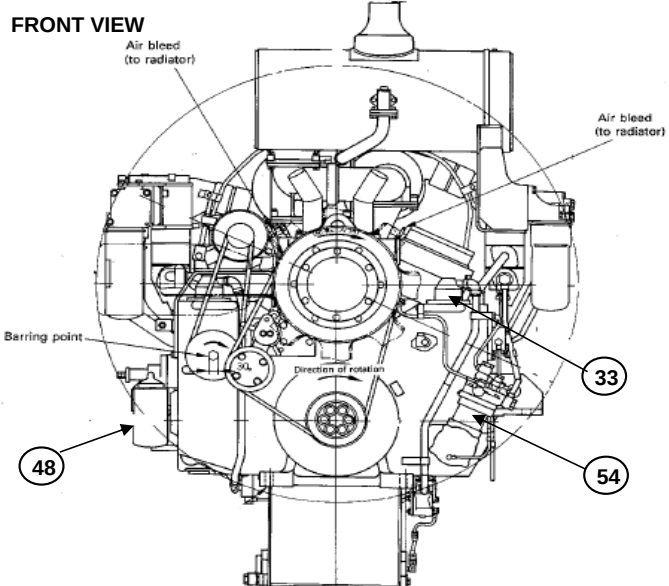
SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
R.H. VIEW 	1	Painting	
	2	Production No.	
	3	Air Crossover Housing	
	4	Turbocharger LH S/N :	
	5	Turbocharger RH S/N :	
	6	Water outlet connection RH	
	7	Lifting Bracket	
	8	Fan belt idler assembly	
	9	After cooler water inlet tube	
	10	Fan belt idler pulley	
	11	Water pump	
	12	Water inlet connection	
	13	Lubricating oil filler tube	
	14	Fuel lift pump	
	15	Lubricating oil drain RH	
	16	Fuel filters	
	17	Fuel injection pump LH S/N :	
L.H. VIEW 	18	Lubricating oil by pass filters	
	19	Lubricating oil transfer tube	
	20	Air intake manifold	
	21	Turbocharger inlet connection	
	22	Turbocharger outlet connection	
	23	After cooler housing	
	24	Cam follower cover	
	25	High pressure fuel supply lines	
	26	Oil pressure sensor	
	27	Starting Motor S/N :	
	28	Lubricating oil drain LH	
	29	Fuel lift pump LH	
	30	Fuel Injection pump RH S/N :	
	31	Vibration damper	
	32	Fuel pump drive	
	33	Crankcase Breather	
	34	Coolant temperature sensor	
	35	Fan hub	
	36	Thermostat housing	
	37	Water vent tubes	

RECEIVING INSPECTION SHEET ENGINE

MACHINE MODEL :	WA 800-3	REPORT NO. :		DOCUMENT NO. :	RC/QR01/EG/01/05
ENGINE MODEL :	SA12V140-1	JOB ORDER NO. :		PUBL. DATE :	
ENGINE NO. :		DELIVERY DATE :		REVISION NO. :	
CUSTOMER NAME :		PAGE :			2 of 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
REAR VIEW 	38	Flywheel housing	
	39	Flywheel	
FRONT VIEW 	40	Engine position sensor (Industrial)	
		Engine speed sensor (G.drive, gen set)	
	41	Engine speed sensor (Industrial)	
	42	Alternator	
		S/N :	
	43	Fan hub	
	44	Air crossover	
	45	Coolant temperature sensor	
	46	Oil pan	
	47	Fan idler tensioner pulley	
	48	Corossion Resistors	
	49	Accessory drive	
	50	Full-flow oil filters	
	51	LH ECM	
		S/N :	
	52	RH ECM	
		S/N :	
	53	Front PTO	
	54	Air Compressor	
	55	Wiring hardness	
	56	Oil Level	

Notes :

Inspected by,

Aproved by,

REV. 00



PLAN REBUILD CENTER

ENGINE ASS'Y DISASSEMBLY CHECKSHEET

KOMATSU ENGINE SA12V140-1

MACHINE MODEL

WA 800-3

WO NO:

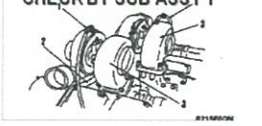
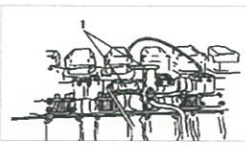
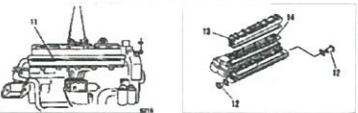
DISASSEMBLY SECTION

DISASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-1,2

JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	1 of 6

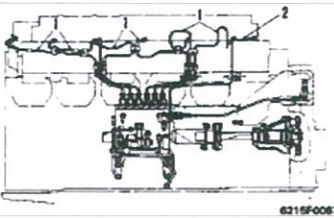
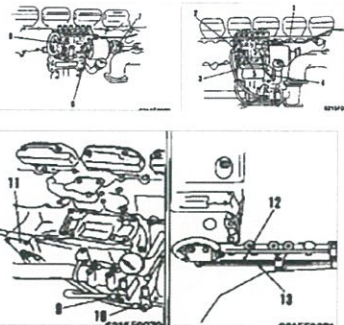
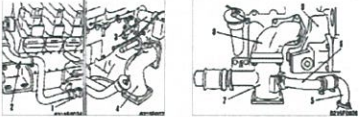
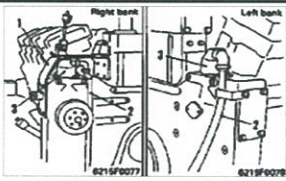
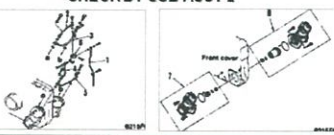
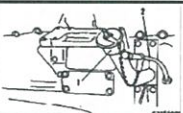
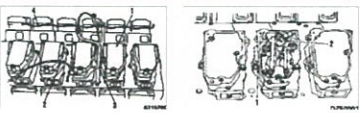
PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOES	REMARKS
1	Engine ass'y	- Setting engine to engine stand Prepare to stable engine stand that will prevent the engine from falling over, the set the engine assembly on the stand and secure it in position. - Drain engine oil from oil pan (the amount of oil differs according to the machine model)				
2	Oil Filter, Fuel Filter and Corrosion Resistor	- Remove and inspect the oil filter housing and lines for damage to pipes, threads, bypass valves, etc - Remove corrosion resistor, tubes and mount. Inspect for damage. - Inspect the fuel filter housing for cracks, pulled threads and damage to fittings.				
3	Air Cleaner Ass'y Muffler Ass'y	- Remove intake connector - Remove and inspect air cleaner if fitted. Check for damage and missing bolts and nuts. - Remove and inspect muffler if fitted. Inspect all nuts and bolts, clamps and adaptors.				
4	Starter Motor Alternator	- Please note if not fitted to engine. - Please note if not fitted to engine.	UPPER LOWER		 CHECK BY SUB ASSY 3	
5	Compressor pipes	Remove air compressor pipes and inspect for damage.				
6	Turbocharger	- Remove Turbocharger, mounting bolts and adapters. - Remove turbo oil line fittings. Check pipes and connections.	LH RH		 CHECK BY SUB ASSY 1	
7	Exhaust Manifold	- Remove and inspect for cracks, pulled threads, snapped studs and missing bolts. - Doing Colour Check inspection joint with QA Officer	FRONT RH CENTRE REAR LH CENTRE REAR			
8	After Cooler	- Check for pulled threads and missing bolts. - Remove aftercooler and check for damage, water leaks, dust entry etc. - Check aftercooler water tubes for cracks, corrosion and any other damage.	LH RH		 CHECK BY SUB ASSY 2	

DISASSEMBLY CHECK SHEET

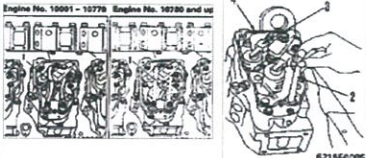
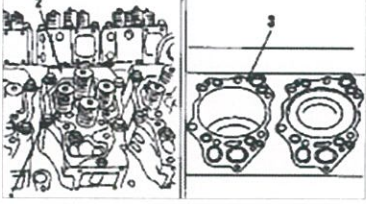
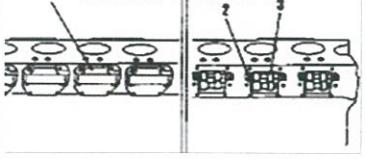
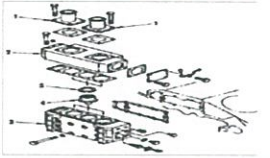
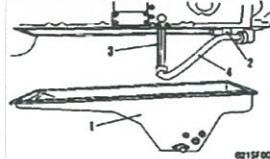
KOMATSU ENGINE 12V140-1,2

JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	2 of 6

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOES	REMARKS														
9	Injection Lines and Clamps	Remove all lines and clamps. Check for wear and damage. <table border="1"> <tr> <td>RH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> </table>	RH	1	2	3	4	5	6	LH	1	2	3	4	5	6				
RH	1	2	3	4	5	6														
LH	1	2	3	4	5	6														
10	Fuel Injection Pump	- Remove pump and mount, and inspect for damage, missing parts and wear. - Check fuel pump oil lines for damage and stripped threads. - Remove and check that fuel pump supply and return lines are in good condition. - Check the fuel pump drive coupling for signs of wear - Check the fuel pump boost tube and clamps for damage.	RH LH		CHECK BY F I P 															
11	Water Pump	- Check for evidence of leakage. - Check for signs of wear and damage.			CHECK BY SUB ASSY 1 															
12	Breather	- Remove and inspect the breather and tube - The water pump is held by two mounting bolts at the front and two mounting bolts at the rear	LH RH																	
13	Air Compressor	- Check for damage and obvious signs of wear. - Remove front compressor with spline shaft - Remove rear compressor with gear	FRONT REAR		CHECK BY SUB ASSY 2 															
14	Oil Cooler	- Check oil cooler housing for cracks and pulled threads. - Check oil cooler for dents and leaks.			 CHECK BY SUB ASSY 2															
15	Oil Filler and Oil Level Gauge	Remove oil filler and oil level gauge, check for dents and cracks																		
16	Rocker Covers and Rocker Housing	Remove Rocker Arm cover and check for cracks and damage to seal face. Note :																		

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-1,2

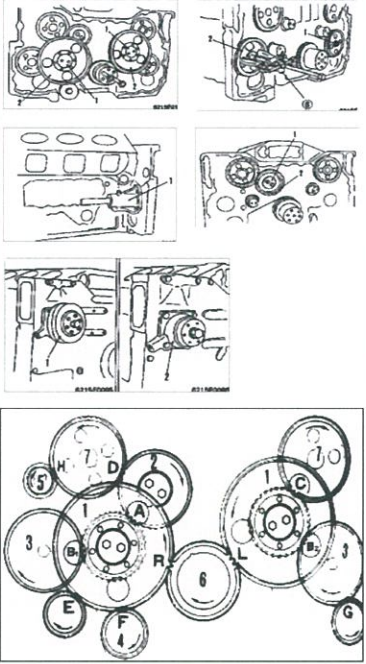
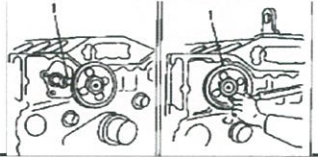
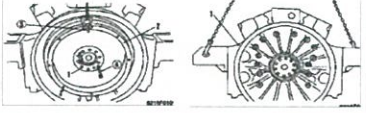
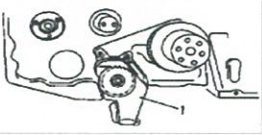
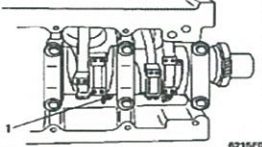
JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	3 of 6

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOES	REMARKS
17	Rocker Arm	- Remove rocker arms and shafts and check for wear and other damage. - Check the contact area on the crossheads are from scuffing and wear. - Check the pushrods for bends and other damage.			CHECK BY SUB ASSY 1 	
18	Cylinder Head	- Check cylinder head bolts for pin punch marks. If there are five marks, renew bolts. - Check the exhaust face condition and bolt holes for pulled threads. - Check the head surface for water marks or damage. - Doing Colour Check inspection joint with QA Officer			CHECK BY SUB ASSY 1 	
19	Cam Followers	- Inspect cam followers for damage to the contact surface of the rollers. - Check condition of covers. The mounting bolts of the cam follower is a special bolt, so keep it in a safe place and be careful not to lose it			CHECK BY SUB ASSY 1 	
20	Thermostat housing Thermostat case	- Disconnect thermostat case and thermostat housing and check condition for cracked and damage. - Check surface contact thermostat				
21	Vibration Damper and Pulleys	- Check vibration damper for damage. - Check pulleys for damage and wear. - Check belt tensioner for wear. - Check front trunnion mount.				
22	Oil Pan	- Remove pan and check for dents and cracks. - Check oil level sensor - Check drain plug for pulled threads - Check oil pick up pipe for cracks in the welds - Check screen for damage.				
23	Front Support and Timing Cover	- Remove and inspect front cover for damage - Check surface condition of seal running area on front shaft.				

DISASSEMBLY CHECK SHEET

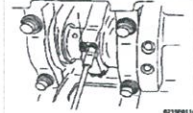
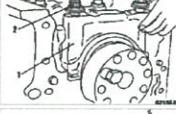
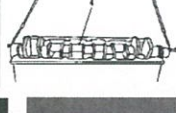
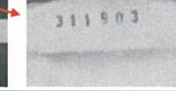
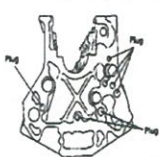
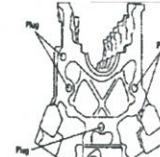
KOMATSU ENGINE 12V140-1,2

JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	4 of 6

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOS	REMARKS																																																																								
24	Timing Gears	- Inspect and record condition of individual gears. Check for chipped teeth, cracks, excessive backlash and end float. - Check shafts and bearings for wear. - Measure thickness of thrust plates for wear. - Check alternator drive. - Check fuel pump drive. - Check accessory drive. - Check water pump drive. Unit : mm <table border="1"> <thead> <tr> <th></th><th>Backlash</th><th>Measured</th><th></th><th>End Play</th><th>Measured</th></tr> </thead> <tbody> <tr> <td>A . A</td><td>0.130 - 0.390</td><td></td><td>1</td><td>0.07 - 0.18</td><td>RH LH</td></tr> <tr> <td>B1. B1</td><td>0.051 - 0.469</td><td></td><td>2</td><td>0.07 - 0.18</td><td></td></tr> <tr> <td>B2. B2</td><td>0.051 - 0.469</td><td></td><td>3</td><td>0.07 - 0.20</td><td>RH LH</td></tr> <tr> <td>C . C</td><td>0.129 - 0.391</td><td></td><td>4</td><td>0.03 - 0.088</td><td></td></tr> <tr> <td>D . D</td><td>0.129 - 0.391</td><td></td><td>5</td><td>0.1 - 0.4</td><td></td></tr> <tr> <td>E . E</td><td>0.052 - 0.481</td><td></td><td>6</td><td>0.14 - 0.32</td><td></td></tr> <tr> <td>F . F</td><td>0.137 - 0.421</td><td></td><td>7</td><td>0.10 - 0.25</td><td>RH LH</td></tr> <tr> <td>G . G</td><td>0.106 - 0.381</td><td></td><td></td><td></td><td></td></tr> <tr> <td>H . H</td><td>0.085 - 0.492</td><td></td><td></td><td></td><td></td></tr> <tr> <td>L . L</td><td>0.141 - 0.425</td><td></td><td></td><td></td><td></td></tr> <tr> <td>R . R</td><td>0.141 - 0.425</td><td></td><td></td><td></td><td></td></tr> </tbody> </table>		Backlash	Measured		End Play	Measured	A . A	0.130 - 0.390		1	0.07 - 0.18	RH LH	B1. B1	0.051 - 0.469		2	0.07 - 0.18		B2. B2	0.051 - 0.469		3	0.07 - 0.20	RH LH	C . C	0.129 - 0.391		4	0.03 - 0.088		D . D	0.129 - 0.391		5	0.1 - 0.4		E . E	0.052 - 0.481		6	0.14 - 0.32		F . F	0.137 - 0.421		7	0.10 - 0.25	RH LH	G . G	0.106 - 0.381					H . H	0.085 - 0.492					L . L	0.141 - 0.425					R . R	0.141 - 0.425							CHECK BY MACHINING 	
	Backlash	Measured		End Play	Measured																																																																									
A . A	0.130 - 0.390		1	0.07 - 0.18	RH LH																																																																									
B1. B1	0.051 - 0.469		2	0.07 - 0.18																																																																										
B2. B2	0.051 - 0.469		3	0.07 - 0.20	RH LH																																																																									
C . C	0.129 - 0.391		4	0.03 - 0.088																																																																										
D . D	0.129 - 0.391		5	0.1 - 0.4																																																																										
E . E	0.052 - 0.481		6	0.14 - 0.32																																																																										
F . F	0.137 - 0.421		7	0.10 - 0.25	RH LH																																																																									
G . G	0.106 - 0.381																																																																													
H . H	0.085 - 0.492																																																																													
L . L	0.141 - 0.425																																																																													
R . R	0.141 - 0.425																																																																													
25	Camshaft	- Check camshaft journals for wear. - Check end float of camshaft. - Check cam surfaces for wear and other damage - Check camshaft for bends.			CHECK BY MACHINING 																																																																									
26	Flywheel and Flywheel Housing	- Check bolts and bolt holes. - Check pilot bearing hole. - Inspect ring gear for chipped and worn teeth. - Check housing for cracks, check bolt holes.																																																																												
27	Oil Pump Ass'y	- Check pump O-rings at inlet and outlet ports. - Check pump for wear and damage.			CHECK BY SUB ASSY 1 																																																																									
	Piston Cooling Nozzles	grooves, skirts and clodge <table border="1"> <thead> <tr> <th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr> </thead> <tbody> <tr> <td>RH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>LH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>		1	2	3	4	5	6	RH							LH																																																													
	1	2	3	4	5	6																																																																								
RH																																																																														
LH																																																																														

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-1,2

JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	5 of 6

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	SALVAGE	REPLACE	SKETCH DRAWING,OR PHOTOES	REMARKS																												
28	Connecting Rods	- Check con rod for cracks and damage. - Check big end bore for scratches or distortion. - Check for damage on mating face of cap. <table><tr><td>RH/LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> - Standard side clereance (End Play) 0.3 ~ 0.454 mm	RH/LH	1	2	3	4	5	6											CHECK BY MACHINING  															
RH/LH	1	2	3	4	5	6																													
29	Pistons	Measure wear on piston skirt, ring grooves and pin bore <table><tr><td>RH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	RH	1	2	3	4	5	6								LH	1	2	3	4	5	6											 	
RH	1	2	3	4	5	6																													
LH	1	2	3	4	5	6																													
30	Crankshaft	- Inspect main and pin journals for scratches, wear, cracks and discoloration. - Check that oil holes are not blocked. - Check fillet area for scratches. - Determine wear by measuring Ø of journals. - Check seal surface condition. - Check cap bolts for number of punch marks. - Check cap fit to block. - Punch S/N engine at the front face of all main bering cap match with S/N engine block				CHECK BY MACHINING      																													
31	Cylinder Liners	- Check for pitting and cracks on the outer circumference of the Cylinder liner - Inspect the inner surface for scoring and corrosion. - Check for pitting damage on seal grooves. - Measure inside Ø, roundness and cylindricity of Cylinder liner <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>RH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>LH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>		1	2	3	4	5	6	RH							LH										 								
	1	2	3	4	5	6																													
RH																																			
LH																																			
32	Cylinder Block	- Check for cracks in the block, especially in the bolt hole part of the cylinder head. - Inspect the lower liner seal area. - Check the blind plug surface for corrosion. - Check all threads. - Check mounting surface of cylinder head for distortion and corrosion. - Check for clogging of oil and water holes. - Doing Colour Check inspection joint with QA Officer				CHECK BY MACHINING  																													

INSPECTION BY

APPROVAL BY

MECHANIC

GL/SPV PRODUCTION

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-1,2

JOB NUMBER		DOCUMENT NO.	QR02-ENG-001
ENGINE MODEL	12V140-1,2	PUBL. DATE	01/08/2008
ENGINE SER. NO		REVISION NO.	04
MACHINE MODEL		PAGE	6 of 6

ATTACHED
OUTSIDE DIAMETER OF CRANKSHAFT PIN JOURNAL AND MAIN JOURNAL
6D170 - 3 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	140,000	139,910	108,000	107,910
0.25 US	139,750	139,660	107,750	107,660
0.50 US	139,500	139,410	107,500	107,410
0.75 US	139,250	139,160	107,250	107,160
1.00 US	139,000	139,910	107,000	106,910
Out of Roundness	0 - 0.010	0,010	0 - 0.010	0,010
Tolerance : - 0.025				

6D140 - 3 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	120,000	119,830	90,000	89,880
0.25 US	119,750	119,630	89,750	89,630
0.50 US	119,500	119,830	89,500	89,380
0.75 US	119,250	119,130	89,250	89,130
1.00 US	119,000	118,880	89,000	88,880
Out of Roundness	0 ~ 0.010	0,020	0 ~ 0.010	0,020
Tolerance : - 0.020				

12V140 - 1, 2 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	148.000 ~ 147.980	147.880 and less	100.000 ~ 99.980	99.880 and less
0.25 US	147.750 ~ 147.730	147.630 and less	99.750 ~ 99.730	99.630 and less
0.50 US	147.500 ~ 147.480	147.380 and less	99.500 ~ 99.480	99.380 and less
0.75 US	147.250 ~ 147.230		99.250 ~ 99.230	99.130 and less
1.00 US	147.000 ~ 146.980		99.000 ~ 98.980	98.880 and less
Out of Roundness	0.010 and less	0.011 and more	0.010 and less	0.011 and more
Tolerance : - 0.020				

OUTSIDE DIAMETER OF CAMSHAFT JOURNAL AND CAM HEIGHT

12V140 - 1,2 Series

Cam Height	Std Size	Tolerance	Repair limit
Intake	55,48	± 0.1	55,08
Exhaust	55,75	± 0.1	55,35
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	65	-0.016 -0.036	

6D140-3 Series

Unit : mm

Cam Height	Std Size	Tolerance	Repair limit
Intake	55,48	± 0.1	55,08
Exhaust	55,75	± 0.1	55,35
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	65	-0.016 -0.036	-

6D170 - 3 Series

Unit : mm

Cam Height	Std Size	Tolerance	Repair limit
Intake	89,8	+0.1081 -0.0919	89,20
Exhaust	90,1	+0.1688 -0.0312	88,80
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	105	-0.042 -0.072	104,88



CYL BLOCK

**MEASUREMENT BEFORE
SALVAGING CHECK SHEET**

KOMATSU ENGINE SA12V140-1

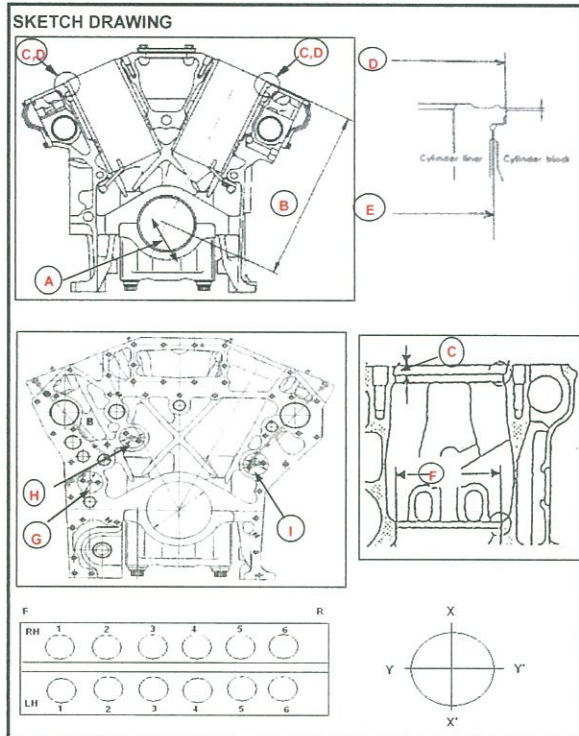
MACHINE MODEL

WA 800-3

DISASSEMBLY SECTION

MEASURING CYLINDER BLOCK BEFORE SALVAGING CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QR03/EG/01/02
ENGINE MODEL	SA12V140 - 1	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	



MECHANIC NAME	Cyl. Block Inspection	
	Cyl. Block assembly	
Standard Size		Unit : mm
DESCRIPTION	STANDARD SIZE	
A	MAIN BEARING BORE	157.000 ~ 157.025
B	CYLINDER BLOCK HEIGHT	471.460 ~ 471.540
	USE OVER SIZE GASKET	471.062 ~ 471.450
C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263
E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230
F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK
G,H,I	TIMING GEAR HOLE	60.000 ~ 60.030

TIGHTENING TORQUE OF MAIN BEARING CAP	1 ST step =	27.5 ~ 30.5 Kgm
	2 ND step =	57.0 ~ 59.0 Kgm
	3 RD step =	90° ~ 120°

VISUAL CHECK	OK	Not OK
CRACK HAS BEEN CHECKED		

MANHOURS UTILIZATION			
	START	FINISH	MAN HOURS
Measuring Process			
TOTAL			

MAIN BEARING BORE

	Main Bore No.		X - X'	Y - Y'	Z - Z'
A	1	A			
		B			
	2	A			
		B			
	3	A			
		B			
	4	A			
		B			
	5	A			
		B			
	6	A			
		B			
	7	A			
		B			

COUNTER BORE DEPTH

C	BEFORE SALVAGING											
	R.H Bank Front	Cyl	Mech				L.H Bank Front	Cyl	Mech			
			X	X'	Y	Y'			X	Y	Y'	Y'
		1						1				
2					2							
3					3							
4					4							
5					5							
6					6							

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/OR03/EG/01/02 2 of 3
ENGINE MODEL	SA12V140 - 1	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263
E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230

INSIDE DIAMETER(AT FLANGE SEAT)									
BEFORE SALVAGING									
D	R.H Bank	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
		6				6			
	L.H Bank	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
		6				6			

TIMING GEAR HOLE(VISUAL CHECK)						
G,H,I	TIMING GEAR HOLES					
	X - X			Y - Y		
	Mac	QA/GL	Spv	Mac	QA/GL	Spv

INSIDE DIAMETER(AT COUNTERBORE)									
BEFORE SALVAGING									
E	RH	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
		6				6			
	LH	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
		6				6			

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/OR03/EG/01/02 3 of 3
ENGINE MODEL	SA12V140 - 1	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK
---	----------------------------------	--------------

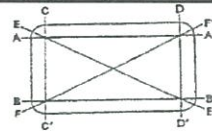
INSIDE DIAMETER (AT O-RING LAND)

F	RH	Cyl	VISUAL CHECK			LH	Cyl	VISUAL CHECK		
			CHECK					CHECK		
			MEC	QA/GL	SPV			MEC	QA/GL	SPV
		1					1			
2				2						
3				3						
4				4						
5				5						
6				6						

Judgement of Distortion (Flatness) Measure thickness between Block & Straight edge.

Flatness	Remedy
within 0.08mm	Correct with surface grinding

Measurement tool: Straight edge, clearance gauge



THICKNESS BETWEEN BLOCK AND STRAIGHT EDGE

RH	Cyl	CHECK			LH	Cyl	CHECK			
		CHECK					CHECK			
		MEC	QA/GL	SPV				MEC	QA/GL	SPV
	1									
	2									
	3									
	4									
	5									
	6									

INSPECTION RESULT :

MEASURED

INSPECTION & APPROVAL BY :

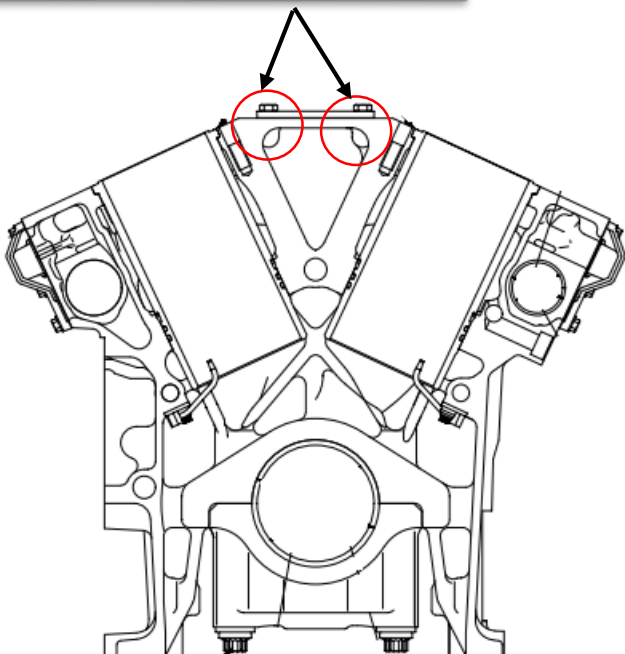
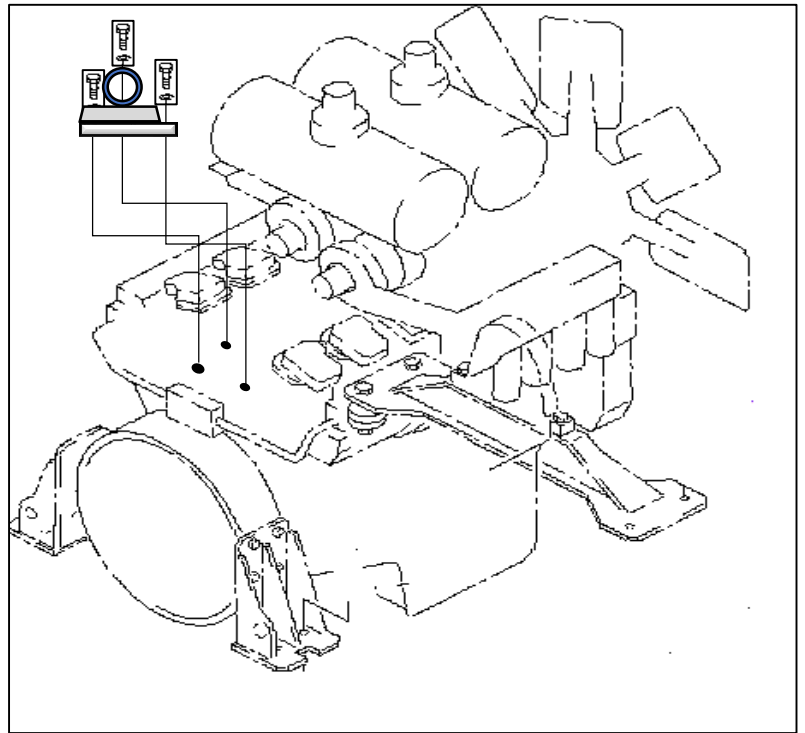
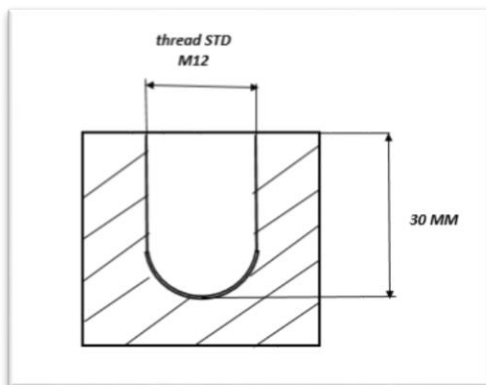
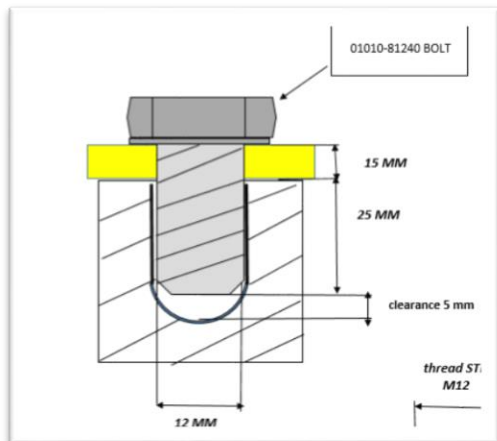
MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

WARNING

PROSES LIFTING ENGINE OR
CYLINDER BLOCK 12V140 SERIES



REV. 03



CRANKSHAFT ASSY MEASURING CHECK SHEET KOMATSU ENGINE SA12V140-1

MACHINE MODEL

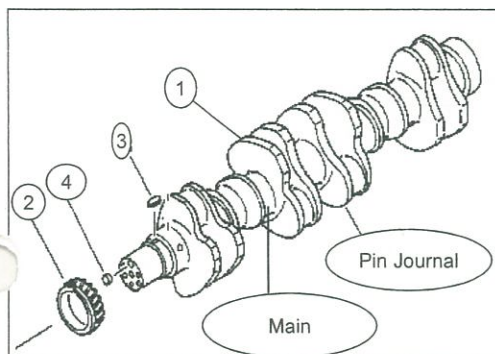
WA 800-3

DISASSEMBLY SECTION

CRANKSHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT ASSY	DOCUMENT NO	RC/QR03/EG/03/02
ENGINE MODEL	12V140-1	PART NUMBER	6215-31-1100/6219-31-1100	PUBL. DATE	07/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 2

SKETCH DRAWING



Physical Data

No.	Part Name	Qty	New	Reuse
1	Crankshaft	1		
2	Gear	1		
3	Key	1		
4	Dowel Pin	1		

MECHANIC NAME

Standard size			Unit : mm	
SIZE	MAIN JOURNAL		PIN JOURNAL	
	STD SIZE	REPAIR LIMIT	STD SIZE	REPAIR LIMIT
STANDARD	148.000 ~ 147.970	147,970	100.000 ~ 99.980	99,880
Under Size 0.25	147.750 ~ 147.730	147,730	99.750 ~ 99.730	99,630
Under Size 0.50	147.500 ~ 147.480	147,480	99.500 ~ 99.480	99,380
ROUNDNESS	0 ~ 0.010		0 ~ 0.010	
STRAIGHTNESS		MAXIMUM = 0.15 mm (TIR)		
FILLET RADIUS CHECK		OK	Not OK	
Main Journal	= 3.0 ~ 3.5 mm			
Pin Journal	= 4.0 ~ 4.4 mm			
VISUAL CHECK		OK	Not OK	
CRACK HAS BEEN CHECKED				
MANHOURS UTILIZATION				
	START	FINISH	MAN HOURS	
Measuring Process				
TOTAL				

INSPECTION SALVAGE PROCESS

CRITICAL INSPECTION			CHECK POINT (DIMENSION)				ROUNDNESS MAX.	
			1			2		
			X - X	Y - Y		X - X		Y - Y
MAIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
	7	MEC						
		QA						

CRANKSHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT ASSY	DOCUMENT NO	RC/QR03/EG/03/02
ENGINE MODEL	12V140-1	PART NUMBER	6215-31-1100/6219-31-1100	PUBL. DATE	07/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

CRITICAL INSPECTION			CHECK POINT (DIMENSION)				ROUNDNESS MAX.	
			1			2		
			X - X	Y - Y		X - X	Y - Y	X - Y
PIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
SRTAIGHTNESS			MAXIMUM = mm / TIR = mm					

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

RC/QA/EG/087

REV. 05



PT.SAPTAINDRA SEJATI

PLAN REBUILD CENTER

CONNECTING ROD

MEASUREMENT CHECK SHEET

KOMATSU ENGINE 12V140E-1

MACHINE MODEL

WO NO:

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CONNECTING ROD	DOCUMENT NO.	
ENGINE MODEL	12V140-1 SERIES	PART NUMBER		PUBL. DATE	02/03/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	

[illegible]

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CONNECTING ROD	DOCUMENT NO.	
ENGINE MODEL	12V140-1 SERIES	PART NUMBER		PUBL. DATE	02/03/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	

CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø SMALL END WITHOUT BEARING					
			X - X'			Y - Y'		
			MAC	QA	SPV	MAC	QA	SPV
1R		A						
2R		A						
3R		A						
4R		A						
5R		A						
6R		A						
1L		A						
2L		A						
3L		A						
4L		A						
5L		A						
6L		A						

CON-ROD NO.	CHECK POINT	Ø SMALL END WITH BEARING					
		X - X'			Y - Y'		
		MAC	QA	SPV	MAC	QA	SPV
1R	A						
2R	A						
3R	A						
4R	A						
5R	A						
6R	A						
1L	A						
2L	A						
3L	A						
4L	A						
5L	A						
6L	A						

[illegible]



PT. SAPTA INDRA SEJATI
PLANT REBUILD CENTER

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CONNECTING ROD	DOCUMENT NO.	
ENGINE MODEL	12V140-1 SERIES	PART NUMBER		PUBL. DATE	02/03/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	

1L		A								
		B								
2L		A								
		B								
3L		A								
		B								
4L		A								
		B								
5L		A								
		B								
6L		A								
		B								

SMALL END AFTER SALVAGE PROCESS (NEW BEARING)

CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø SMALL END WITHOUT BEARING					
			X - X'			Y - Y'		
			MAC	QA	SPV	MAC	QA	SPV
1R		A						
		B						
2R		A						
		B						
3R		A						
		B						
4R		A						
		B						
5R		A						
		B						
6R		A						
		B						

SMALL END AFTER SALVAGE PROCESS (NEW BEARING)

CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø SMALL END WITHOUT BEARING					
			X - X'			Y - Y'		
			MAC	QA	SPV	MAC	QA	SPV
1L		A						
		B						
2L		A						
		B						
3L		A						
		B						
4L		A						
		B						
5L		A						
		B						
6L		A						
		B						



PT. SAPTA INDRA SEJATI
PLANT REBUILD CENTER

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CONNECTING ROD	DOCUMENT NO.	
ENGINE MODEL	12V140-1 SERIES	PART NUMBER		PUBL. DATE	02/03/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	

SMALL END AFTER SALVAGE PROCESS BY OPERATOR

CON-ROD NO.	CHECK POINT	Ø SMALL END WITH BEARING					
		X - X'			Y - Y'		
		MAC	QA	SPV	MAC	QA	SPV
1R	A						
	B						
2R	A						
	B						
3R	A						
	B						
4R	A						
	B						
5R	A						
	B						
6R	A						
	B						
1L	A						
	B						
2L	A						
	B						
3L	A						
	B						
4L	A						
	B						
5L	A						
	B						
6L	A						
	B						

CON ROD BENDING

CON-ROD NO.	BEND		TWIST	
	Indicator Reading	Actual Bending	Indicator Reading	Actual Twist
1R				
2R				
3R				
4R				
5R				
6R				
1L				
2L				
3L				
4L				
5L				
6L				

BEND AND TWIST

Measure with formula :

$$\frac{\text{Distance between hole}}{\text{Width seat metal}} \times 1 \text{ Reading}$$

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CONNECTING ROD	DOCUMENT NO.	
ENGINE MODEL	12V140-1 SERIES	PART NUMBER		PUBL. DATE	02/03/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	

INSPECTION RESULT :

INSPECTION & APPROVAL BY :

PROD. SUPERVISOR (SPV)

REV. 03



CAMSHAFT RH

PROCESS AND MEASURING

CHECK SHEET

KOMATSU ENGINE SA12V140-1

MACHINE MODEL

WA 800-3

DISASSEMBLY SECTION

CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CAMSHAFT RH	DOCUMENT NO.	
ENGINE MODEL	12V140-1	PART NUMBER		PUBL. DATE	RC/QR03/EG/07/01
ENGINE S/N		START DATE/TIME		REVISION NO.	03
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
	START	FINISH	MAN HOURS
Machining Process			
Measuring Process			
TOTAL			



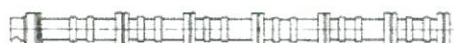
Right bank Camshaft



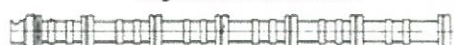
Left bank Camshaft

Unit : mm

STANDARD		REPAIR LIMIT	No		CAMSHAFT JOURNAL			CAM LUBE INTAKE			CAM LUBE EXHAUST		
					MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV
12V140	Intake = 55.58 ~ 55.38	55,08	1L	X									
	Exhaust = 55.65 ~ 55.85	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		2L	X									
				Y									
6D140	Intake = 55.58 ~ 55.38	55,08	3L	X									
	Exhaust = 55.85 ~ 55.65	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		4L	X									
				Y									
6D170-3	Intake = 91.240 ~ 91.040	89,20	5L	X									
	Exhaust = 90.268 ~ 90.068	88,80		Y									
	Cam Journal = 104.958 ~ 104.928	104,88	6L	X									
				Y									
6D170-2	Intake = 62.414 ~ 62.214	61,37	7L	X									
	Exhaust = 60.678 ~ 60.478	59,64		Y									
	Cam Journal = 71.92 ~ 71.89	71,86	1R	X									
				Y									
DESCRIPTION	OK	Not OK	2R	X									
				Y									
CRACK HAS BEEN CHECK			3R	X									
				Y									
			4R	X									
				Y									
			5R	X									
				Y									
			6R	X									
				Y									
			7R	X									
				Y									



Right bank Camshaft



Left bank Camshaft

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 03



CAMSHAFT LH

PROCESS AND MEASURING

CHECK SHEET

KOMATSU ENGINE SA12V140-1

MACHINE MODEL

WA 800-3

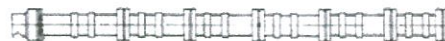
DISASSEMBLY SECTION

CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CAMSHAFT LH	DOCUMENT NO.	
ENGINE MODEL	12V140-1	PART NUMBER		PUBL. DATE	RC/QR03/EG/07/01
ENGINE S/N		START DATE/TIME		REVISION NO.	03
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
	START	FINISH	MAN HOURS
Machining Process			
Measuring Process			
TOTAL			



Right bank Camshaft



Left bank Camshaft

Unit : mm

STANDARD		REPAIR LIMIT	No		CAMSHAFT JOURNAL			CAM LUBE INTAKE			CAM LUBE EXHAUST		
					MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV
12V140	Intake = 55.58 ~ 55.38	55,08	1L	X									
	Exhaust = 55.65 ~ 55.85	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		2L	X									
				Y									
6D140	Intake = 55.58 ~ 55.38	55,08	3L	X									
	Exhaust = 55.85 ~ 55.65	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		4L	X									
				Y									
6D170-3	Intake = 91.240 ~ 91.040	89,20	5L	X									
	Exhaust = 90.268 ~ 90.068	88,80		Y									
	Cam Journal = 104.958 ~ 104.928	104,88	6L	X									
				Y									
6D170-2	Intake = 62.414 ~ 62.214	61,37	7L	X									
	Exhaust = 60.678 ~ 60.478	59,64		Y									
	Cam Journal = 71.92 ~ 71.89	71,86	1R	X									
				Y									
DESCRIPTION		OK	Not OK	2R	X								
				Y									
CRACK HAS BEEN CHECK				3R	X								
				Y									
				4R	X								
				Y									
				5R	X								
				Y									
				6R	X								
				Y									
				7R	X								
				Y									



Right bank Camshaft



Left bank Camshaft

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

RC/QA/EG/088

REV. 05



PT.SAPTAINDRA SEJATI

PLAN REBUILD CENTER

PIN PISTON

MEASUREMENT CHECK SHEET

KOMATSU ENGINE 12V140E-1

MACHINE MODEL

WO NO:

PISTON PIN MEASURING AND POLISHING CHECK SHEET

JOB NUMBER		PART NAME	PISTON PIN	DOCUMENT NO.	RC/QA/EG/088
ENGINE MODEL	12V140-1,2	PART NUMBER		PUBL. DATE	07/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

STANDARD (unit mm)	Clearance limit
12V140 Series = Ø 52.000 - 51.994	0,10

QUANTITY	
----------	--

INSPECTION MEASURING PROCESS											
CYL.NO.	CHECK POSITION	1			2			3			Clearance actual
		MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
RIGHT BANK	1	X - X'									
		Y - Y'									
	2	X - X'									
		Y - Y'									
	3	X - X'									
		Y - Y'									
	4	X - X'									
		Y - Y'									
	5	X - X'									
		Y - Y'									
	6	X - X'									
		Y - Y'									
LEFT BANK	1	X - X'									
		Y - Y'									
	2	X - X'									
		Y - Y'									
	3	X - X'									
		Y - Y'									
	4	X - X'									
		Y - Y'									
	5	X - X'									
		Y - Y'									
	6	X - X'									
		Y - Y'									

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 01



PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYLINDER BLOCK INSPECTION CHECK SHEET

KOMATSU ENGINE 12V140 SERIES

MACHINE MODEL

WO NO. :

ENGINE ASS'Y SECTION

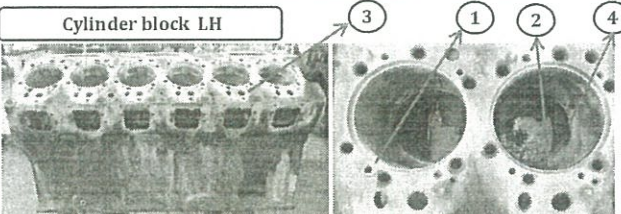
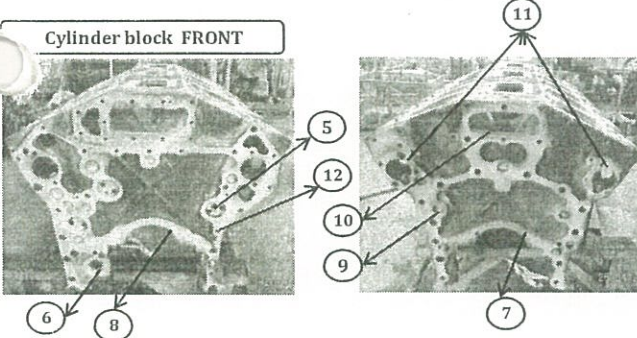
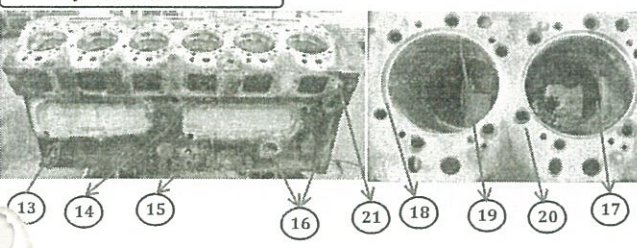
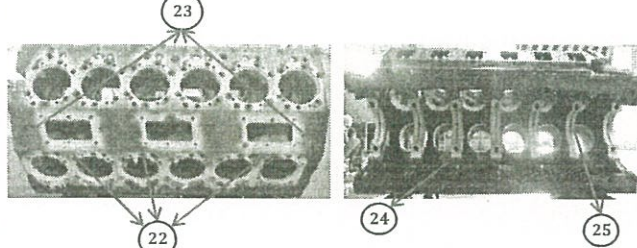
CYLINDER BLOCK INSPECTION SHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6219-21-1100	DOCUMENT NO.	ENG-CB-12V140/3-2-001
ENGINE MODEL	SAA12V140 SERIES	INSPECTION DATE		PUBL. DATE	5-Aug-14
ENGINE NO.		WO NUMBER		REVISION NO.	03
CUSTOMER NAME				PAGE	2 OF 3

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOS	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
Cylinder block LH 	1	Counter bore surface area			
	2	Water jacket wall area			
	3	Cylinder block upper surface			
	4	O ring liner area			
 Contoh pengecekan thread pada cyl block dengan bolt		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD 72 THREAD AND 7 PLUG
		# All thread on camfollower side			
		# All thread cyl head&Top surface LH cyl block			
		# All thread on (common rail side cyl block)			
Cylinder block FRONT 	5	Timing gear hole area			
	6	Oil pump contact area			
	7	Thrust bearing surface			
	8	Main bore front surface			
	9	Oil and water hole gallery surface area			
	10	Blind cover water area			
	11	Camshaft hole surface rear area RH LH			
	12	Remove all plug on oil and water gallery			
		All Thread condition			
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD 48 FRONT SIDE, 40 THREAD REAR SIDE
		# All thread timing gear shaft & Camshaft			
	# All thread front cover mounting				
Cylinder block RH 	13	Oil cooler hole area			
	14	Oil adapter hole area			
	15	Regulator hole area			
	16	Water hole area			
	17	O ring liner area			
	18	Counter bore surface area			
	19	Water jacket wall area			
	20	Cylinder block upper surface			
	21	All Thread condition			
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD 112 THREAD
		# All thread Camfollower & Oil cooler area			
	# All thread oil adapter and under oil cooler				
	# All thread cyl head & all thread on surface RH				
	22	Cylinder block blind cover area			
	23	Retap thread on front and rear hanger			
	24	Thread mounting piston cooling nozzle			
	25	All Main metal upper seat area			
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD UPPER 26 THREAD, LOWER 72 THREAD
		# All thread cyl block hanger and blind cover area			
	# All thread oil pan, main cap & cooling nozzle mounting				

Notes :

1. Semua thread yang tertutup dengan tanah (soil) harus dilakukan retap dan cleaning oleh man power disassy.

Checked by,

Inspected by,

Aproved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)

CRANKSHAFT INSPECTION SHEET ENGINE

MACHINE MODEL		CRANKSHAFT P/N	6215-21-1100	DOCUMENT NO.	ENG-CS-12V140/1-2-001
ENGINE MODEL	SA12V140-1,-2	INSPECTION DATE		PUBL. DATE	13-Feb-15
ENGINE NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			GOOD	NO GOOD	
Crankshfat assy	1	Fillet radius Main and Pin journal no. 1 from pitting & Corrosion			
	2	Fillet radius Main and Pin journal no. 2 front and rear side from pitting &			
	3	Fillet radius Main and Pin journal no. 3 front and rear side from pitting and corrosion			
	4	Fillet radius Main and Pin journal no.4 from pitting and corrosion			
	5	Fillet radius Main and Pin journal no.5 from pitting and corrosion			
	6	Fillet radius Main and Pin journal no.6 from pitting and corrosion			
	7	Fillet radius main journal no.7 from pitting and corrosion			
	8	Check condition front & rear seal contact area from wear and fretting			
	9	Check crankgear condition from pitting and chipping. (front gear for engine 12V140 and front-rear gear for engine 170)			
	10	Check condition web, counter weight from chipping and pitting.			
	11	Check condition all thread crankshaft front and rear. Make sure engine S/N was marking on rear surface area			
Note: Engine serial Number position marking code, location on Rear Pilot bearing area (free contact area with Flywheel).					
Notes :					

Checked by,

Inspected by,

Approved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)



PLANT REBUILD CENTER

ENGINE ASS'Y

SHORT BLOCK

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-1

MACHINE MODEL

WO NO:

SHORT BLOCK SECTION

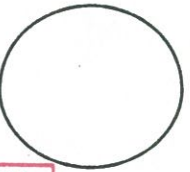
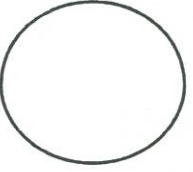
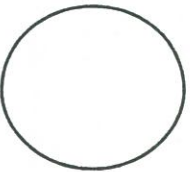
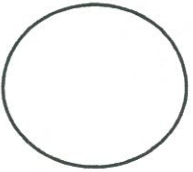
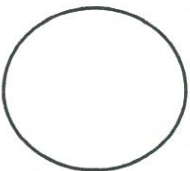
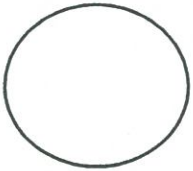
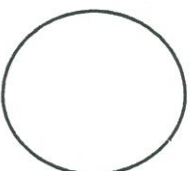
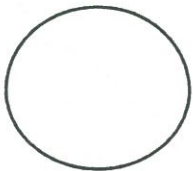
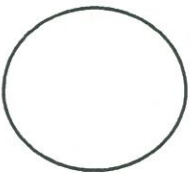
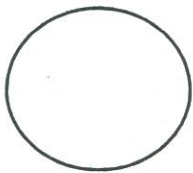
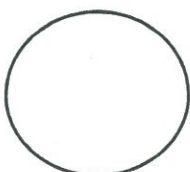
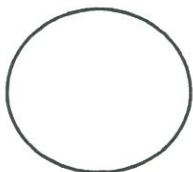
MEASUREMENT LINER PROTRUSION EACH CYLINDER

FRONT ENGINE



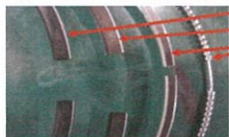
LH

RH



CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC.DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)																																								
					MEC	QA	SPV																																									
3	Crank Shaft Cont.	End Play	Measure crankshaft end play with dial gauge Act : mm • Tolerance : After tightening bolts, check that shaft turns smoothly. Turning torque act : kgm	0.14~0.32 mm (Shop Manual Page. 13-048) Max. 1.5 Kgm (Shop Manual Page. 13-048)				  																																								
4	Piston and Connect. Rods	Piston	When assembling a new con rod, mark the cylinder number with an electric pen. Do not make a stamped mark.					 To 2 O   																																								
		piston ring	Set up each ring with the stamped mark facing up. Note the positioning of the ring gap. For the oil ring, take out the expander and install it onto the piston, then install the oil ring. Confirm that the expander is properly fitted to the groove, and end gaps of oil ring and expander are 180° apart. Align the casting mark EX-F on the piston with the chamfered side of the con. rod. numbers in the same direction, insert the piston pin to assemble the piston connecting rod																																													
		Snap ring	Confirm that the snap ring which secures the piston pin is properly fitted to the groove. Check that the connecting rod moves smoothly																																													
		Con. Rod Bearing	- Check size of main bearings STD 0.25 0.50 - Match the number of con. rods and Check that there is no dirt at the rear face of the bearing before assembling. <table><tr><th>RH Bank</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>Upper</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Lower</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><th>LH Bank</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>Upper</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Lower</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> Do not coat the rear surface of the bearing with oil.	RH Bank	1	2	3		4	5	6	Upper							Lower							LH Bank	1	2	3	4	5	6	Upper							Lower							STD or US 0.25 US 0.50	
	RH Bank	1	2	3	4	5	6																																									
	Upper																																															
	Lower																																															
	LH Bank	1	2	3	4	5	6																																									
	Upper																																															
	Lower																																															
Inserting Piston and Con. Rod	Check that the hole in the upper bearing and the oil hole in the con. rod are aligned. Set the cylinder block on its side so that the assembly does not enter the cylinder suddenly when inserted. Coat the cylinder liner with engine oil (SAE 15W-40) uniformly around the whole circumference by hand Check that number of piston and connecting rod assembly match cylinder number. Coat connecting rod lower bearing with engine oil, spreading over whole part with finger. Conrod bolts are to have oil (SAE 15W-40) on the threads and under the washers. 1 st step: (tighten in order shown on diagram) Act.: kgm 2 nd step: Repeat above step Act.: kgm 3 rd step: Act.: ° 4 th recheck torque bolt. Act.: kgm Measure con rod side clearance . <table><tr><th>Cyl.No</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>RH&LH Bank</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> • Connecting rod bolts must be marked (punch) after tighten overhaul	Cyl.No	1	2	3	4	5	6	RH&LH Bank							CRITICAL ITEM CHECK Name : Name : QA INSPECTOR PROD. SUPERVISOR																																
	Cyl.No	1	2	3	4	5	6																																									
	RH&LH Bank																																															
	Big End Torque	Make sure that piston position is in correct position. (see from top position) * Top Piston with deep notch is sitting to intake side * Top Piston with deep notch is sitting to exhaust side <table><tr><th>Cyl.no</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><td>L bank</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>R bank</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Cyl.no	1	2	3	4	5	6	L bank							R bank							7.0 ~ 8.0 Kgm 7.0 ~ 8.0 Kgm Turn 90° +30° 13 ± 1 Kgm (Shop Manual Page. 13-053) 0.30~0.454 mm (Shop Manual Page. 13-053)																								
		Cyl.no	1	2	3	4	5	6																																								
		L bank																																														
R bank																																																
Piston Position																																																

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1

JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
ENGINE MODEL	SA12V140-1	PUBLIC DATE	
ENGINE SERIAL NO.		REVISION NO.	04
MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)																									
					MEC	QA	SPV																										
4	Piston and Con Rod cont.	Piston Cooling Nozzle	Check that the piston cooling nozzle is not clogged and that the angle is correct. Use thread tightener (LT-2) on the mounting bolt. Act.: kgm Please marking on the table <table border="1"> <thead> <tr> <th>Part name</th> <th>Liner</th> <th>Piston</th> <th>Pin Piston</th> <th>Ring Piston</th> <th>Conrod</th> <th>Conrod Bolt</th> </tr> </thead> <tbody> <tr> <td>New</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Used</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>	Part name	Liner	Piston	Pin Piston	Ring Piston	Conrod	Conrod Bolt	New							Used							5.5 ± 2.0 Kgm (Shop Manual Page. 13-053)				  tread tightener LT-2) on the mounting bolt thread				
Part name	Liner	Piston	Pin Piston	Ring Piston	Conrod	Conrod Bolt																											
New																																	
Used																																	
15	Cylinder Head		Check that there is no dust or dirt inside the cylinder and on mounting surface of cylinder head. When installing the gasket, check that the grommet has been fitted and has not come out of position. Grommets The grommets in the head gasket may become deformed by swelling, so do not coat them with gasket sealant. O/S head gasket If oversize head gaskets are to be fitted, the block must be stamped (O/S) <table border="1"> <thead> <tr> <th>STD</th> <th>0.25</th> <th>0.50</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> </tr> </tbody> </table> Bolts The head bolts must have no dirt in the threads or under the washer, or in the threaded holes in the block. A tap can be used to clean the holes. Coat the thread and washer of the bolts with oil. Cyl. Head Torque <table border="1"> <thead> <tr> <th>Step</th> <th>Act :</th> <th></th> <th></th> </tr> </thead> <tbody> <tr> <td>1st step:</td> <td></td> <td>Kgm</td> <td>15.0 ± 1.0 Kgm</td> </tr> <tr> <td>2nd step:</td> <td></td> <td>Kgm</td> <td>22.0 ± 0.5 Kgm</td> </tr> <tr> <td>3rd step:</td> <td></td> <td>°</td> <td>turn 90° +30°</td> </tr> <tr> <td>For bolt 7 :</td> <td></td> <td>Kgm</td> <td>6.0 ~ 7.5 Kgm (Shop Manual Page. 13-067)</td> </tr> </tbody> </table> After tightening, make a punch mark on the bolt head. If the bolt has five marks, do not use it. Make sure Installation heat label at cylinder head No.6 LH & RH Act : °	STD	0.25	0.50				Step	Act :			1 st step:		Kgm	15.0 ± 1.0 Kgm	2 nd step:		Kgm	22.0 ± 0.5 Kgm	3 rd step:		°	turn 90° +30°	For bolt 7 :		Kgm	6.0 ~ 7.5 Kgm (Shop Manual Page. 13-067)	CRITICAL ITEM CHECK STD or OVER SIZE Name : QA INSPECTOR PROD. SUPERVISOR			  CODE FOR OVERSIZE GASKET  
STD	0.25	0.50																															
Step	Act :																																
1 st step:		Kgm	15.0 ± 1.0 Kgm																														
2 nd step:		Kgm	22.0 ± 0.5 Kgm																														
3 rd step:		°	turn 90° +30°																														
For bolt 7 :		Kgm	6.0 ~ 7.5 Kgm (Shop Manual Page. 13-067)																														



PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYL BLOCK

MEASUREMENT AND INSPECTION CHECKSHEET

KOMATSU ENGINE

SA12V140-1 SERIES

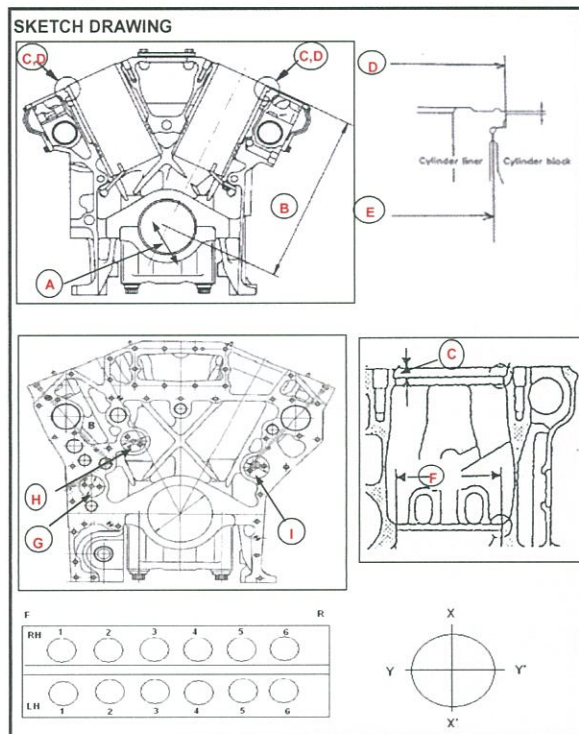
MACHINE MODEL

WO NO:

SHORT BLOCK SECTION

MEASURING CYLINDER BLOCK INSPECTION CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	



MECHANIC NAME		Cyl. Block Inspection
		Cyl. Block assembly
Standard Size		Unit : mm
DESCRIPTION	STANDARD SIZE	
A	MAIN BEARING BORE	157.000 ~ 157.025
B	CYLINDER BLOCK HEIGHT	471.460 ~ 471.540
	USE OVER SIZE GASKET	471.062 ~ 471.450
C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263
E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230
F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK
G,H,I	TIMING GEAR HOLE	60.000 ~ 60.030

TIGHTENING TORQUE OF MAIN BEARING CAP	1 ST step =	27.5 ~ 30.5 Kgm
	2 ND step =	57.0 ~ 59.0 Kgm
	3 RD step =	90° ~ 120°

VISUAL CHECK	OK	Not OK
CRACK HAS BEEN CHECKED		

MANHOURS UTILIZATION			
	START	FINISH	MAN HOURS
Measuring Process			
TOTAL			

MAIN BEARING BORE

Main Bore No.		X - X'	Y - Y'	Z - Z'
A	1	A		
		B		
	2	A		
		B		
	3	A		
		B		
	4	A		
		B		
	5	A		
		B		
	6	A		
		B		
	7	A		
		B		

COUNTER BORE DEPTH

C	BEFORE MACHINING											
	R.H Bank Front	Cyl	Mech				L.H Bank Front	Cyl	Mech			
			X	X'	Y	Y'			X	Y	Y'	Y'
		1						1				
		2						2				
		3						3				
		4						4				
		5						5				
6					6							

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	2 of 4 RC/QAVEG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263

AFTER MACHINING														
C	R.H Bank	Cyl	X			X'			Y			Y'		
			Mech	QA	Spv	Mech	QA	Spv	Mech	QA	Spv	Mech	QA	Spv
		1												
		2												
		3												
		4												
		5												
		6												
	L.H Bank	Cyl	X			X'			Y			Y'		
			Mech	QA	Spv	Mech	QA	Spv	Mech	QA	Spv	Mech	QA	Spv
		1												
		2												
		3												
		4												
		5												
		6												

INSIDE DIAMETER(AT FLANGE SEAT)

D	R.H Bank	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
	6				6				
	L.H Bank	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
	6				6				

TIMING GEAR HOLE(VISUAL CHECK)

G,H,I	TIMING GEAR HOLES					
	X - X			Y - Y		
	Mac	QA/GL	Spv	Mac	QA/GL	Spv

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	3 of 4 RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START.DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230
F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK

INSIDE DIAMETER(AT COUNTERBORE)

E	RH	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
		3				3			
		4				4			
		5				5			
	6			6					
	LH	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
			MEC	QA/GL	SPV		MEC	QA/GL	SPV
		1				1			
		2				2			
3				3					
4				4					
5				5					
6			6						

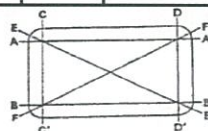
INSIDE DIAMETER (AT O-RING LAND)

F	RH	Cyl	VISUAL CHECK			LH	Cyl	VISUAL CHECK		
			CHECK					CHECK		
			MEC	QA/GL	SPV			MEC	QA/GL	SPV
		1					1			
2				2						
3				3						
4				4						
5				5						
6				6						

Judgement of Distortion (Flatness) : Measure thickness between Block & Straight edge.

Flatness	Remedy
within 0.08mm	Correct with surface grinding

Measurement tool : Straight edge, clearance gauge



THICKNESS BETWEEN BLOCK AND STRAIGHT EDGE

RH	Cyl	CHECK			LH	Cyl	CHECK			
		CHECK					CHECK			
		MEC	QA/GL	SPV				MEC	QA/GL	SPV
	1									
	2									
	3									
	4									
	5									
	6									

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	4 of 4 RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START.DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

INSPECTION RESULT :		
MEASURED MECHANIC	INSPECTION & APPROVAL BY : QUALITY ASSURANCE (QA) GL/SPV PRODUCTION	



PLANT REBUILD CENTER

CRANKSHAFT

MEASUREMENT AND

INSPECTION CHECK SHEET

KOMATSU ENGINE 12V140-1

MACHINE MODEL

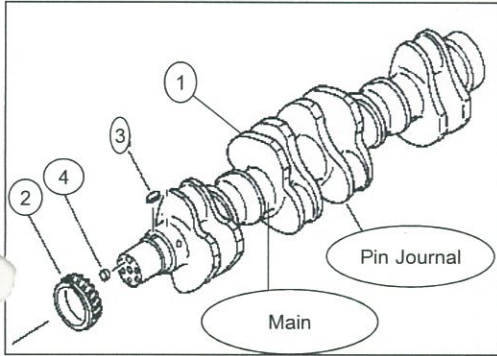
WO NO:

SHORTBLOCK SECTION

CRANKSHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT ASSY	DOCUMENT NO	RC/QR03/EG/03/02
ENGINE MODEL	12V140-1	PART NUMBER	6215-31-1100/6219-31-1100	PUBL. DATE	07/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL	WA 800-3	FINISH DATE / TIME		PAGE	1 of 2

SKETCH DRAWING



Physical Data

No.	Part Name	Qty	New	Reuse
1	Crankshaft	1		
2	Gear	1		
3	Key	1		
4	Dowel Pin	1		

MECHANIC NAME

Standard size			Unit : mm	
SIZE	MAIN JOURNAL		PIN JOURNAL	
	STD SIZE	REPAIR LIMIT	STD SIZE	REPAIR LIMIT
STANDARD	148.000 ~ 147.970	147,970	100.000 ~ 99.980	99,880
Under Size 0.25	147.750 ~ 147.730	147,730	99.750 ~ 99.730	99,630
Under Size 0.50	147.500 ~ 147.480	147,480	99.500 ~ 99.480	99,380
ROUNDNESS	0 ~ 0.010		0 ~ 0.010	
STRAIGHTNESS		MAXIMUM = 0.15 mm (TIR)		
FILLET RADIUS CHECK		OK		Not OK
Main Journal	= 3.0 ~ 3.5 mm			
Pin Journal	= 4.0 ~ 4.4 mm			
VISUAL CHECK		OK		Not OK
CRACK HAS BEEN CHECKED				
MANHOURS UTILIZATION				
	START	FINISH		MAN HOURS
Measuring Process				
TOTAL				

INSPECTION SALVAGE PROCESS

CRITICAL INSPECTION			CHECK POINT (DIMENSION)				ROUNDNESS MAX.	
			1		2			
			X - X	Y - Y	X - X	Y - Y		X - Y
MAIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
	7	MEC						
		QA						

CRANKSHAFT MEASURING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT ASSY	DOCUMENT NO	RC/QR03/EG/03/02
ENGINE MODEL	12V140-1	PART NUMBER	6215-31-1100/6219-31-1100	PUBL. DATE	07/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	04
MACHINE MODEL	WA 800-3	FINISH DATE / TIME		PAGE	2 of 2

CRITICAL INSPECTION			CHECK POINT (DIMENSION)					ROUNDNESS MAX.
			1			2		
			X - X	Y - Y		X - X	Y - Y	
PIN JOURNAL	1	MEC						
		QA						
	2	MEC						
		QA						
	3	MEC						
		QA						
	4	MEC						
		QA						
	5	MEC						
		QA						
	6	MEC						
		QA						
SRTAIGHTNESS		MAXIMUM = mm / TIR = mm						

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

CYLINDER BLOCK WASHING CHECKSHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6219-21-1100	DOCUMENT NO.	ENG-CB-12V140/3-2-001
ENGINE MODEL	12V140 SERIES	INSPECTION DATE		PUBL. DATE	5-Aug-14
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOS	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
Cylinder block Front and Rear	1	Remove U plug on front cyl block area and rear area. (Main gallery, piston cooling gallery, water gallery line)			
	2	Check seat of u plug from chipped and pitted.			
	3	Clean seat of U plug from corrosive and dirty oil.			
	4	Clean seat of big u lug (2 pcs) on front side cyl block. make sure free from corrosive and dirty.			
	5	Clean all of gallery (22mm) use nylon brush with adapter and make sure clean on all gallery. Any 5 gallery (1 main gallery, 2 piston cooling gallery, 2 water cooling gallery).			
	6	Clean front side and rear side contact on cyl block use wire brush. make sure none abnormal flat on all contact surface and no corrosion.			
Cylinder block Lower surface	7	Clean lubricating hole on main bore cylinder block (7 holes). use nylon brush 8mm			
	8	Clean hole of piston cooling nozzle and thread mounting every cylinder (12 holes) use nylon brush 8mm			
	9	Clean all thread for main cap bolt use nylon brush and make sure all of thread free from corrosion and dirty oil (14 holes).			
Cyl block oil cooler and camfollower thread	10	Clean and retap mounting bolt oil pan on cyl block and make sure no dirty and corrosion on all thread. And clean all contact surface on cyl block lower area.			
	11	Clean lubricating hole on oil cooler front and rear use nylon brush 22mm.			
	12	Clean lubricating hole on cyl block for oil adapter area use nylon brush 22mm.			
Cyl block surface and thread of surface	13	Clean mounting hole for lubricating camfollower and mounting thread use nylon brush 12mm (every cyl have 8 holes).			
	14	Retap all bolt mounting cyl head thread on top surface use special TAP BOLT, then clean use nylon brush 22mm with washing chemical.			
	15	Clean lubricating small hole for cyl head on top surface use nylon brush 8mm and make sure none dirty oil on lubricating hole every cylinder.			
	16	Clean all lubricating hole (main gallery, piston cooling gallery, camshaft lubricating gallery etc) use new engine oil and. And use special tool oil spray gun. Don't spray oil on water cooling gallery.			

* Tool A is pin dowel new for check Go or no Go.

Notes :

Checked by,

Inspected by,

Approved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)

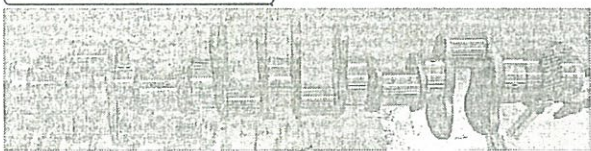
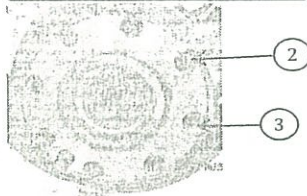
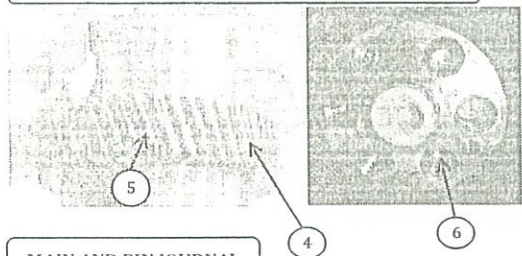
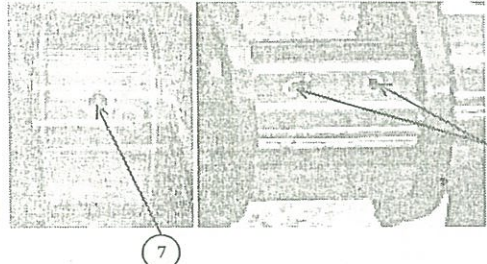
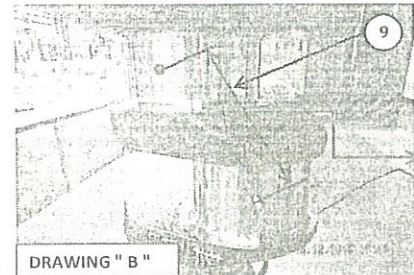
CRANKSHAFT WASHING CHECKSHEET ENGINE

MACHINE MODEL		CRANKSHAFT P/N		DOCUMENT NO.	ENG-CR-12V140/3-2-001
ENGINE MODEL	12V140 SERIES	WASHING DATE		PUBL. DATE	29-DES-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊕ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
CRANKSHAFT ASSY  1 REAR THREAD CONDITION  2 3 CRANKGEAR AND FRONT THREAD CONDITION  4 5 6 FRONT DAMPER BOLT  7 MAIN AND PIN JOURNAL  7 8 DRAWING " B "  9 DISTANCE " L "	1	Check condition crankshaft assy from corrosion and abnormality scratch on all main and pin journal before washing.			
	2	Recheck all thread condition, use FLYWHEEL BOLT for check condition of thread.			
	3	Clean all thread use nylon brush (22 mm) and washing chemical until none dirty oil on all thread.			
	4	Recheck all front thread condition use FRONT DAMPER BOLT for check all thread condition.			
	5	Clean gear from corrosion and dirty, use nylon brush or wire brush and washing chemical.			
	6	Clean all front thread use nylon brush (22 mm) use washing chemical.			
	7	Clean all main journal use nylon brush (8 mm) with extension. Until lubricating hole clean and none dirty oil.			
	8	Clean all pin journal use nylon brush (8 mm) with extension. Until lubricating hole clean and none dirty oil.			
	9	Clean Pin journal lubricating hole cross until main journal lubricating hole (distance L), for distance see drawing " B "			
* Lubricating hole on pin journal for engine 12V140 series have 2 holes, and all engine 6D140, 6D170, 6D125 series have 1 hole.					
Notes :					

Checked by,

Inspected by,

Aproved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)

CONNECTING ROD WASHING CHECKSHEET

ENGINE

MACHINE MODEL		CONNECTING ROD P/N		DOCUMENT NO.	ENG-CR-6D12V140/3-2-00
ENGINE MODEL	12V140 SERIES	WASHING DATE		PUBL. DATE	29-Dec-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

[illegible]

Checked by,

Inspected by,

Aproved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)

MACHINE MODEL		CONNECTING ROD P/N		DOCUMENT NO.	ENG-CR-6D12V140/3-2-00
ENGINE MODEL	12V140 SERIES	WASHING DATE		PUBL. DATE	29-Dec-15
ENGINE SERIAL NO.		WO NUMBER		REVISION NO.	00
CUSTOMER NAME				PAGE	1 OF 1

⊗ : NONE

[illegible]

Aproved by,

(PROD.SUPERVISOR)

REV.01



PLANT REBUILD CENTER

ENGINE ASS'Y

MAIN LINE

ASSEMBLY CHECK SHEET

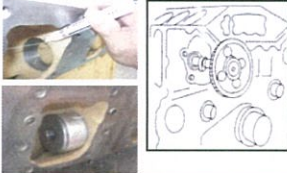
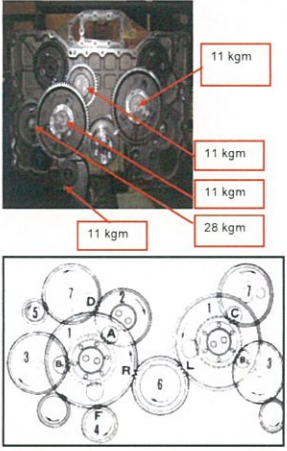
KOMATSU ENGINE 12V140-1

MACHINE MODEL

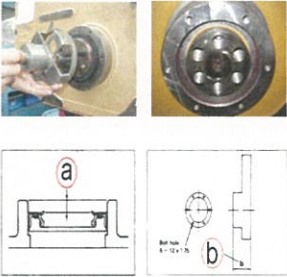
WO NO:

MAIN ASS'Y SECTION

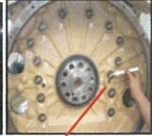
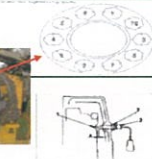
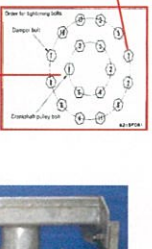
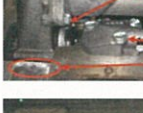
ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)	
					MEC	QA/GL		
5	Camshaft	Assembly	Check camshaft bushes are correctly installed in block, and oil supply holes are aligned. When installing the camshaft assembly (2), rotate it to protect the cam bushing from damage. Match the marks on the camshaft and gear. • NyLock bolt of camshaft plate: Act : Kgm	9.0 ± 3.5 Kgm (Shop Manual Page. 13-054)				
6	Timing Gears	FIP Drive Gear	Install fuel pump drives. • Fuel injection pump drive gear drive gear nut: Act : Kgm	25.5 ± 2.5 Kgm (Shop Manual Page. 00-12)				
		Oil Pump	Fit O-ring and install oil pump assembly. • Oil pump bolt: Act : Kgm	11.5 ± 1.0 Kgm (Shop Manual Page. 00-12)				
		Small Idler	Install small idler gear 2.					
		Large Idler	Fit plates with the out mark facing the outside and • tighten mounting bolts Act : Kgm	$10 \sim 12.5$ Kgm (Shop Manual Page. 13-056)				
			Make sure that all match marks are correctly aligned					
			Tighten mounting bolts double gear RH & LH $\varnothing 10$ mm 12 pcs. Act : Kgm Tighten mounting bolts double gear RH & LH $\varnothing 8$ mm 12 pcs. Act : Kgm	6.70 ± 0.7 kgm (Shop Manual Page. 00-12) 3.2 ± 0.3 kgm (Shop Manual Page. 00-12)				
		Backlash & End play	Measure the backlash and end play of all gears and record in table below:	(Shop Manual Page. 13-058)				

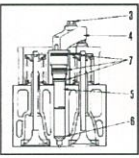
Timing Gears Marked	Backlash (mm)	Measured	SIGN			No.	Check Items	End Play	SIGN			sured
			MEC	QA	SPV				MEC	QA	SPV	
A . A	0.130 ~ 0.390					1	Idler Gear (large)	0.07 ~ 0.18				
B ₁ .B ₁	0.051 ~ 0.469					2	Idler Gear (small)	0.07 ~ 0.18				
B ₂ .B ₂	0.051 ~ 0.469					3	FIP drive gear	0.07 ~ 0.20				
C . C	0.129 ~ 0.391					4	Oil pump drive	0.03 ~ 0.088				
D . D	0.129 ~ 0.391					5	Acc. Drive gear	0.1 ~ 0.4				
F . F	0.137 ~ 0.421					6	Crank Shaft gear	0.14 ~ 0.32				
L . L	0.141 ~ 0.425					7	Cam shaft gear	0.10 ~ 0.25				
R . R	0.141 ~ 0.425											

7	Gear Case Cover	Seal lip	clean of al dirt and dust on the contact surface of the front seal	(G2-L1) STD or OVER SIZE (Shop Manual Page. 13-059)	
		Front seal	Check front seal specification S/W STD OS S = Single Lip Seal W = Double Lip Seal		
		Measure dimension of portion (b) of front cover			
		Press in the front seal uniformly so that it is not at an angle. Than use table to press-fitting depth (a) oil seal Use installation tool to obtain correct position.			
		Dimension of potion b	Bolt hole 6= 12x1.75	Press-fitting depth a for oil seal	
		109 mm	No	13-14 mm	
		114 mm	Yes	18-19 mm	

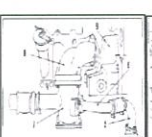
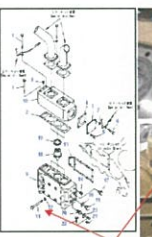
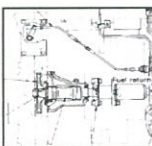
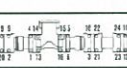
ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
8	Flywheel Housing	Rear seal	Check rear seal specification SW STD OS S = Single Lip Seal W = Double Lip Seal Press in the rear seal uniformly so that it is not at an angle, and check that the spring has not come out. Press-fitting depth of seal: Act : mm	STD or OVER SIZE 18~19 mm (Shop Manual Page. 13-060-1)			    
			Use installation tool to obtain correct position. Be careful not to damage the seal when installing the housing.				
			Torque bolts in order shown. (Coat with oil).	35 ± 5 Kgm			
			1 st step: Act : Kgm 2 nd step: Act : Kgm 3 rd step: Act : Kgm	loosen bolts completely. 42 ± 2 Kgm (Shop Manual Page. 13-060-2)			
9	Flywheel		Before assembly, coat bolts with engine oil (SAE15W-40).				 
			Flywheel mounting bolts:				
			1 st step: Act : Kgm	8.0~12.0 Kgm			
			2 nd step: Act : Kgm	28.0~32.0 Kgm			
10	Cam Follower	assembly	3 rd step: Act : Kgm	53.0~57.0 Kgm (Shop Manual Page. 13-060)			 
			Install speed sensor and adjust clearance Act : mm				
			Before installing the cam follower assembly, wash the oil hole of the mounting bolt, then immerse whole assembly in engine oil (SAE 15W-40).				
			Use LG-6 liquid gasket on the cover face.				
11	Damper & Pulley		Torque the cam follower mounting bolt : Act : Kgm	5.25±0.75 Kgm (Shop Manual Page. 13-065)			 
			Torque the cover plate mounting bolts Act : Kgm	4.0 ± 4.5Kgm (Shop Manual Page. 13-065)			
			Check that the vibration damper has been tested or renewed before installing.				
			Coat the thread and washer of the bolts with engine oil (SAE15W-40)				
12	Oil Pump	Gear Drive	Tighten damper mounting bolts to Pulley. Act : Kgm	11.0 ± 1.0 Kgm (Shop Manual Page. 13-064)			 
			Tighten the crank pulley mounting bolts in the order shown.				
			1 st step: Act : Kgm	5.5 ~ 9.5 Kgm			
			2 nd step: Act : Kgm	23.0~27.0 Kgm			
13	Oil Pan		3 rd step: Act : Kgm	74.0~78.0 Kgm (Shop Manual Page. 13-064)			 
			Check oil pump ass'y completely				
			Fill in Engine oil (SAE 15W-40)				
			Rotate oil pump smoothly				
12	Oil Pump	Strainer	Fit O-Ring and mounting the oil pump				 
			Tighten mounting bolts Act : Kgm	11.5±1 Kgm			
			End Play of oil pump Act : mm	0.03~0.088 mm (Shop Manual Page. 13-058)			
			Recheck plug of main relief valve (retighten)				
12	Oil Pump	Strainer	Make sure that strainer clean (inner & outer)				 
			Fit O-Ring				
			Tighten mounting bolts Act : Kgm	11.5±1 Kgm			
			Mounting the bracket Act : kgm	6.70 ±0.7 Kgm (Shop Manual Page. 00-12)			
13	Oil Pan		Check that the oil pan is not crack & clean				 
			Arrange the gasket to the oil pan				
			When tightening, start from center and work in order to the out side.				
			Tighten mounting bolts Act : Kgm	6.70 ±0.7 Kgm (Shop Manual Page. 00-12)			

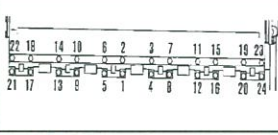
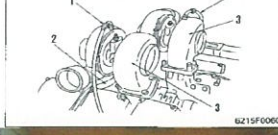
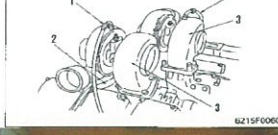
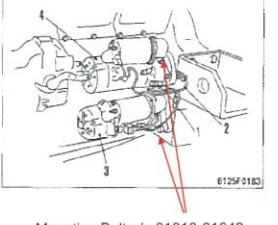
ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
14	Oil Cooler		- Check that there is no dust or dirt inside the oil cooler and no crack - Coat the oil cooler gasket with gasket sealant (three bond) and install the oil cooler ass'y to the cyl. block. - Tighten mounting bolts : Act : Kgm	6.7 0 ± 0.7 kgm			 
15	Cross Heads		Fit cross heads and adjust so that when the screw has touched the valve stem it advanced 20°. Cross head adjusting screws go to exhaust manifold side of head. Locknut Torque : Act : Kgm Check that the push rod is fitted in the cam follower properly.	6.0 ± 0.6 Kgm (Shop Manual Page. 13-068)			 
16	Rocker Arm Housing		When installing, be careful not to let the O-ring come off or get caught. <i>Note:</i> after engine serial no.10780, no O-ring fitted.				 
		Housing	Torque mounting bolts.(M 10) Act : Kgm	6.70 ±0.7 Kgm			
		Housing	Torque mounting bolts(M12). Act : Kgm	11.5 ± 1.0 Kgm (Shop Manual Page. 00-12)			
17	Rocker Arm Lever		Confirm that the ball of the adjusting screw is fitted to the socket of the push rod before tightening bolt. Tightening bolt. Act : Kgm	10.0 ± 0.5 Kgm (Shop Manual Page. 13-069)			 
		Valve gap	Adjust valve gap to: Intake: Act : mm	0.43 mm			
		Firing Order:	Exhaust: Act : mm R1-L1-R5-L5-R3-L3-R6-L6-R2-L2-R4-L4	0.80 mm (Shop Manual Page. 12-002)			
18	Nozzle Holder		Confirm that there is no dust in the nozzle holder sleeve. Fit O-rings to nozzle and lower seat seal, and coat with engine oil. Fit gasket, align nozzle holder with mounting hole of inlet connector and insert nozzle holder. Install one end of the holder to the rocker arm housing, lay on the bolt head and tighten with the bolt - Holder mounting bolt (engines serial no.up to 10306) Act : Kgm - Holder mounting bolt (engines serial no.10307 and up) - Inlet connector Act : Kgm - Locknut Act : Kgm	6.70 ±0.7 Kgm 3.5 ~ 4.0 Kgm 3.0 ~ 3.5 Kgm (Shop Manual Page. 13-070)			   
19	Rocker Arm Cover		- Tighten rocker cover mounting bolts - Fixed type (New) Act : Kgm - Full floating (Old) Act : Kgm	2.5 Kgm (Reman Manual Page. 8.52) 0.5 Kgm (Reman Manual Page. 8.52)			 
20	Oil Level gauge	Oil filler	Fit the O-ring and tighten the nut at the root of the level gauge guide to secure it in position. Act : Kgm	5.25 ±0.75Kgm (Shop Manual Page. 13-071)			
21	Air Compressor		When installing air compressor, coat spline shaft with grease. Tighten nut Act : Kgm	17.5 ± 2.5 Kgm (Shop Manual Page. 13-071)			 
22	Breather		Tighten mounting bolt Act : Kgm <i>Note :</i> some breathers are fitted with an O-ring and some are not. The alternator bracket is on the right bank only.	2.75 ±0.25Kgm (Shop Manual Page. 13-071)			

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)						
					MEC	QA/GL							
23	Water Pump	Water pump	Tighten two mounting bolts for the water pump from the front, and two from the rear. <table><tr><th colspan="2">water pump</th></tr><tr><th>NEW</th><th>REUSE</th></tr><tr><td></td><td></td></tr></table> • TightenTorque : Act : Kgm	water pump		NEW	REUSE			11.5 ± 1.0 Kgm (Shop Manual Page. 00-12)			
		water pump											
		NEW	REUSE										
Water pump	Confirm that the tubes no dust on inside. Fit the gasket and install the tubes and tighten bolts Act : Kgm	11.5 ± 1.0 Kgm (Shop Manual Page. 00-12)			 								
Tubes	Fit the O Ring and install the by pass tubes then connect master joint and secure clamp				 								
Thermo-stat Housing	• Press fit seal to thermostat case • Make sure that the seal is not at an angle when press fitting • Assemble thermostat in thermostat case Fit gasket, the install thermostat case to thermostat housing and tighten mounting bolts Act : Kgm Fit gasket and install thermostat case assembly to cylinder block and tighten mounting bolts Act : Kgm	7 kgm (Reman Manual Page. 8.52) 11.5 ± 1.0 Kgm (Shop Manual Page. 00-12)			 								
24	Fuel Injection Pump		When installing the injection pump, first rotate the crankshaft in the direction of rotation until the timing mark on the engine pulley is aligned.				 Bolt p/n.01011-81220 Fuel Injection Pump Mechanical Governor  Fuel Injection Pump Electrical Governor   						
			Then align the stamped line "a" on the flange and line "b" on the flange of the pump.										
			Set the timing of the fuel pump for the right hand bank first. Then move crankshaft in direction of rotation and set left hand bank. • Mounting bolt Act : Kgm	7 kgm (Reman Manual Page. 8.52)									
			• Coupling lock bolt M 12 x 1.75 Act : Kgm	9.5 kgm (Reman Manual Page. 8.52)									
			• M 14 x 2.0 (electrical) Act : Kgm	20 kgm (Reman Manual Page. 8.52)									
			• Plate bolt Act : Kgm	9.5 kgm (Reman Manual Page. 8.52)									
			• Fuel injection pipe Act : Kgm	2.4± 0.1Kgm (Shop Manual Page. 13-075)									
25	Exhaust Manifold		Tighten the mounting bolts in the order shown in the diagram. Coat the threads with grease before tightening. Torque : Act : Kgm	5.0 ± 0.5 Kgm (Shop Manual Page. 13-075)			 						

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140-1	JOB NO.		DOCUMENT NO	RC/QR04/EG/01/02
	ENGINE MODEL	SA12V140-1	PUBLIC DATE	
	ENGINE SERIAL NO.		REVISION NO.	04
	MACHINE MODEL	WA 800-3	PAGE	

NO	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
26	Intake Manifold	Intake Manifold After Cooler	To prevent the electrical air intake heater from falling off, install a guide bar in the mounting bolt hole of the intake manifold when installing. • Tighten the bolts in the order shown in the diagram. Act : Kgm Fit gasket and install after cooler assembly to intake manifold with tighten mounting bolts Torque : Act : Kgm	6.75±0.75 Kgm (Shop Manual Page. 13-076) 6.75±0.75 Kgm (Shop Manual Page. 13-076)			  
27	Turbo-charger		When installing the turbocharger mounting bolts, install each connection, then align the center and tighten fully. Torque : Act : Kgm	5.0 ± 0.5 Kgm (Shop Manual Page. 13-077)			
28	Exhaust Brake		Install exhaust brake assembly together with bracket Fit gasket and install connector then tighten clamp * See picture beside!!!				
29	Starting Motor		Fit gasket and install upper starting motor and than tighten mounting bolt. Torque : Act : Kgm Fit gasket, install lower starting motor and than tighten mounting bolt. Torque : Act : Kgm Fit bracket and install 2 wiring connectors	20 kgm (Reman Manual Page. 8.52) 20 kgm (Reman Manual Page. 8.52)			 Mounting Bolt p/n.01010-61640

Carry out a visual inspection of the engine prior to dyno testing

* Critical point must be marked by stamp

UNIT MODEL	INJECTION TIMING (deg.)
D475A-2	30
D475A-3	9/12
HD785-3	18
HD785-5	18
HD785-7	CRI
WA800-2	30
WA800-3M	30
WA800-3E	-
WA900-3M	29
WA900-3E	-

Reference : Factory Standard Reference

NOTES :

ASSEMBLY BY

INSPECTION & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PLANT REBUILD CENTER

ENGINE ASS'Y

ENGINE PERFORMANCE

TEST CHECKSHEET

KOMATSU ENGINE 12V140-1

MACHINE MODEL

WA 800-3

JOB NO:

ENGINE ASS'Y SECTION



ENGINE PERFORMANCE TEST REPORT

DOCUMENT NO.	QR05-ENG-009
PUBLICATION DATE	04/07/2009
REVISION NO.	02
PAGE	1 of 2

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SA12V140-1	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	WA800-3M (Mech.Gov)	TEST BY	

RUNNING TEST				
No	STANDARD (MANUAL)		ACTUAL TESTED	
	RPM	HP	RPM	HP
1	1040	68		
2	1250	141		
3	1550	348		
4	1800	603		
5	2000	846		

MEASUREMENT BLOWBY PRESSURE AT RATED SPEED (WITH MANUAL TEST)				
	STANDARD PRESSURE		ACTUAL	
	MAXIMUM PRESSURE	150 mmH ₂ O(1.5 Kpa)	1st	
	PERMISSIBLE VALUE	300 mmH ₂ O(2.9 Kpa)	2nd	
	ALTERNATOR TEST :			
	STANDARD MAXIMUM		ACTUAL LOW IDLE	ACTUAL HIGH IDLE
	LOW IDLE:	HIGH IDLE:		

MAINTENANCE STANDARD								
SPEED (Rpm)		OIL		WATER		BOOST PRESS (mmHg)	BLOW BY (mmH2O)	EXH TEMP (°C)
		Press. (Kg/cm²)	Temp (°C)	Press. (Kg/cm²)	Temp (°C)			
Low idle	680 ~ 730	Min. 1.5	90 ~ 110	-	70~90	Min. 1000 at rated HP permissible 800	Max 150 permissible 300	Max 650 at rated HP and torque
High idle	2300 ~ 2350	3.5 ~ 5.0						
POWER (HP) / RPM		827 ~ 878 / 2000 ± 50						
TORQUE (Kgm) / RPM		376 ~ 398 / 1400 ± 100						
EXH.TEMP DIFFERENT		Maximum 10 celcius degree RH and LH Bank						

CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

ENGINE PERFORMANCE TEST REPORT

DOCUMENT NO.	QR05-ENG-009
PUBLICATION DATE	04/07/2009
REVISION NO.	02
PAGE	1 of 2

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SA12V140-1	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	WA800-3M (Mech.Gov)	TEST BY	

QUALITY PROBLEM

DESCRIPTION	REMARKS
WHAT (PROBLEM NAME)	
WHERE (PROBLEM LOCATION)	
WHEN (PROBLEM DATE)	
WHO (INSPECTED BY)	
HOW (PROBLEM EXPLAIN)	

CORRECTIVE ACTION:

PREVENTIVE ACTION :

REMARKS :

[illegible]



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

ENGINE PERFORMANCE TEST REPORT	DOCUMENT NO.	QR05-ENG-009
	PUBLICATION DATE	04/07/2009
	REVISION NO.	02
	PAGE	1 of 2

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SA12V140-1	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	WA800-3M (Mech.Gov)	TEST BY	

RECHECK VALVE CLEARANCE AFTER LOAD

Adjustment valve clearance	* Penentuan posisi top cylinder engine harus dicek dan disetujui oleh QA.		
	Step and method adjustment of value clearance following to the shop manual attached.		
	Firing Order: R1-L1-R5-L5-R3-L3-R6-L6-R2-L2-R4-L4.		
	Valve clearance:		
Rocker Arm cover	Intake valve : 0.43	Act : mm	
	Exhaust valve : 0.80	Act : mm	
	Tighten locknut to secure adjustment screw.		
	Locknut: 6.0 ± 0.6 kgm	Act : kgm	
Rocker Arm cover	Fit O ring and install rocker arm cover.		
	Fixed type (New) : 2.5 kgm (Reman manual page 8.52)		
Rocker Arm cover	Full floating (Old) : 0.5 kgm (Reman manual page 8.52)		
	Mounting bolt actual :	Act : kgm	

INSPECTION BY

INSPECTION & APPROVAL BY

MECHANIC

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)